

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H},$

$a = 13, b = 6$

$$f(p|\mathcal{D}) = \frac{p^{12}(1-p)^5}{B(13,6)}$$

